

Perceptron Learning Law with Decision Region - Practical Report

Aim

To implement the Perceptron Learning Law using Python and visualize its decision region graphically.

Objectives

1. To understand the working of a perceptron.
2. To implement perceptron learning using Python.
3. To classify binary input data.
4. To visualize the decision region graphically.

Theory

A perceptron is the simplest type of artificial neural network used for binary classification problems. It takes input values, multiplies them with weights, adds bias, and passes the result through an activation function. The perceptron learns by updating weights using the perceptron learning rule until correct classification is achieved.

Formulas

Perceptron Learning Rule:

$$w(\text{new}) = w(\text{old}) + \eta(t - y)x$$

Decision Boundary Equation:

$$w_1x_1 + w_2x_2 + b = 0$$

Algorithm

1. Initialize weights and bias to zero.
2. Provide training data and target outputs.
3. Calculate net input.
4. Apply activation function.
5. Compute error.
6. Update weights and bias.
7. Repeat until all epochs complete.
8. Plot decision region graphically.

Efficient Python Code

```
import numpy as np
import matplotlib.pyplot as plt

X = np.array([[0,0],[0,1],[1,0],[1,1]])
y = np.array([0,0,0,1])

w = np.zeros(2)
b = 0
lr = 0.1
for _ in range(10):
```

```

    for i in range(len(X)):
        y_pred = 1 if np.dot(X[i], w) + b >= 0 else 0
        error = y[i] - y_pred
        w += lr * error * X[i]
        b += lr * error

xx, yy = np.meshgrid(np.arange(-0.5,1.5,0.01),
                     np.arange(-0.5,1.5,0.01))

Z = np.where(np.dot(np.c_[xx.ravel(), yy.ravel()], w) + b >= 0, 1, 0)
Z = Z.reshape(xx.shape)

plt.contourf(xx, yy, Z, alpha=0.3, cmap='coolwarm')
plt.scatter(X[:,0], X[:,1], c=y, s=100, cmap='coolwarm')

plt.title("Perceptron Decision Region")
plt.xlabel("X1")
plt.ylabel("X2")
plt.grid()
plt.show()

```

Code Explanation

- NumPy is used for numerical calculations.
- Matplotlib is used for graphical visualization.
- X contains input data and y contains target outputs.
- Weights and bias are initialized to zero.
- The perceptron updates weights using the learning rule.
- Meshgrid is used to create decision regions.
- contourf() plots the decision boundary graphically.

Expected Output

The program displays:

- Final trained weights and bias
- Graphical decision region
- Classified input points

Instructions to Run (.py file) in Anaconda

1. Open Anaconda Navigator.
2. Launch Spyder or VS Code.
3. Create a new Python file.
4. Paste the code into the file.
5. Save the file with .py extension.
6. Install libraries using:
pip install numpy matplotlib
7. Run the program using the Run button or F5 key.

Instructions to Run (.ipynb file) in Jupyter Notebook

1. Open Anaconda Navigator.
2. Launch Jupyter Notebook.
3. Open the .ipynb notebook file.
4. Run each cell using Shift + Enter.
5. The final cell displays the graphical output.

Advantages

- Simple and easy to implement
- Fast training process
- Useful for binary classification problems

Limitations

- Works only for linearly separable data
- Cannot solve XOR-type problems directly

Outcome

The perceptron model was successfully trained and the decision region was visualized graphically using Python.

Conclusion

The practical successfully demonstrates the perceptron learning law and classification of binary data. The perceptron updates weights iteratively and creates a linear decision boundary to separate classes.

Important Viva Questions with Answers

Question	Answer
What is a perceptron?	A perceptron is a simple neural network used for binary classification.
What is the perceptron learning rule?	It is a rule used to update weights based on prediction error.
What is the role of bias?	Bias shifts the decision boundary.
What is a decision boundary?	It is a line separating different classes.
Which activation function is used?	Binary step activation function.
Why do we use meshgrid?	To visualize the decision region graphically.
What are epochs?	Epochs represent complete passes through training data.
What is learning rate?	It controls the amount of weight update.
Can perceptron solve XOR problem?	No, because XOR is not linearly separable.